



Healthy Eating *for Life*

Physicians
Committee
for Responsible Medicine
PCRM.org



**Food Choices for Cancer
Prevention and Survival**

For the first time ever, more than 2 million cancer cases are projected to be diagnosed in the United States over the course of just one year.¹ Cancer is one of the leading causes of death worldwide, and there is an urgent need for a new direction in battling this disease. That's why the Physicians Committee for Responsible Medicine has created materials specifically for cancer prevention and survival.

The Physicians Committee has two cancer-fighting goals. First, we aim to make cancer prevention a top priority. Second, and just as important, we want to improve survival after cancer has been diagnosed by offering comprehensive information about the role of dietary factors in keeping people healthy.

The Physicians Committee provides cooking classes, books, television and video programs, web-based information, brochures, and other educational materials on cancer prevention and survival. To find a class in your community, visit PCRM.org/FFL.

Thank you for your interest and support.

Neal Barnard, MD



Physician, researcher, and author Neal D. Barnard, MD, is one of America's leading advocates for health, nutrition, and higher standards in research. Dr. Barnard is the founder and president of the Physicians Committee for Responsible Medicine. He also initiated Food for Life, a program dedicated to cancer prevention and survival.



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HEALTHY EATING FOR LIFE is not intended as individual medical advice. Always discuss any diet change with your personal physician. In some cases, diet changes may alter your need for medication. People who follow a vegan diet should be sure to include a source of vitamin B12 in their daily routine, such as fortified cereals, fortified soy milk, or any common multiple vitamin.

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Food Choices for Cancer Prevention and Survival

Research shows that about 40% of cancers in the United States are due to factors that have been identified and can potentially be controlled.²

Not only do we have the potential to prevent many cancers, we can also improve the survival rates of people who have cancer.

Cancer starts when one cell begins to multiply out of control. It begins to expand into a lump that can invade healthy tissues and spread to other parts of the body. But this deadly disease can often be prevented, and when it occurs, it can often be stopped.

Studies looking at diet and cancer show that much of our risk for colon, breast, and prostate cancer, among other types, is due to dietary factors.^{3,4}

The link between diet and cancer is not new. In January 1892, *Scientific American* printed the observation that “cancer is most frequent among those branches of the human race where carnivorous habits prevail.” Numerous research studies have since shown that cancer is much more common in populations consuming diets rich in fatty foods, particularly meat and dairy products, and much less common in countries with diets rich in grains, vegetables, fruits, and legumes. One reason is that foods affect the action of hormones in the body. They also affect the strength of the immune system. While fruits and vegetables contain a variety of vitamins, minerals, antioxidants, and phytochemicals that protect the body, research shows that animal products contain potentially carcinogenic compounds that may contribute to increased cancer risk.



Established or Suspected Obesity-Related Cancers⁴

Breast (postmenopausal)
Prostate (advanced)
Pancreas
Esophagus (adenocarcinoma)
Gastric Cardia (adenocarcinoma)
Endometrium
Colon and Rectum
Liver
Gallbladder
Kidney (renal cell)
Non-Hodgkin's Lymphoma
Multiple Myeloma
Leukemia
Stomach (men)
Ovary
Uterus
Cervix

Building Your Strength Against Cancer

Some dietary changes have a preventive effect for many types of cancer. Boosting your intake of vitamin-rich vegetables and fruits, for example, strengthens your immune system and helps knock out cancer cells. To help prevent cancer, it is also smart to avoid meats, dairy products, and fried foods. Choosing fiber-rich legumes, grains, vegetables, and fruits helps keep many types of cancer at bay. Plant foods also contain a wide variety of cancer-fighting substances called phytochemicals.

These facts all point to choosing a vegan diet to help prevent cancer and improve cancer survival. Studies of those following a plant-based diet show that death rates from cancer are only about one-half to three-quarters of those of the general population. Breast cancer rates are dramatically lower in countries such as China and Japan, where diets are typically based on rice, vegetables, and bean products, with very little use of meat, dairy products, or oily foods. When people from those countries adopt a Western, meat-based diet, their breast cancer rates soar.

Are you ready to start enjoying the powerful benefits of a vegan diet? See “The Three-Step Way to Go Vegan” on page 13.

Defending Your DNA From Harm

Let’s look at the steps we can take to build our general defenses. Oxygen is essential to life. But as oxygen is used in the body, some of the oxygen molecules become very unstable. These unstable molecules, called free radicals, can attack cell membranes and even damage the DNA (our genetic code) in the nucleus of the cell. Damage to DNA is the beginning of cancer.

Fortunately, the foods we eat can help protect our bodies. Antioxidants, including vitamin C, vitamin E, beta-carotene, selenium, lycopene, and others, can neutralize the damaging effects of oxygen. These powerful natural chemicals come to us in vegetables, fruits, grains, and beans. People who include fruits and vegetables in their daily diets have lower rates of many forms of cancer.

To step up the antioxidant power of your menus, try a baked sweet potato, Easy Bean Salad, or baked corn chips with Colorful Corn Salsa (recipes on pages 18 and 19).

Even with vegetables and fruits in the diet, damage to the cells’ DNA will occasionally occur, so the body has built-in repair machinery. Fixing damaged DNA, which could otherwise cause cancer, requires a B vitamin called folate, which is found in dark green leafy vegetables, fruits, peas, and beans. Evidence suggests that folate may be particularly important in preventing HPV-related cervical cancer.⁵

The Dietary Reference Intake for folate for adult women and men is 400 micrograms per day and increases to 600 micrograms per day for pregnant women. Beans and vegetables are rich in folate. Asparagus, black beans, black-eyed peas, chickpeas, lentils, pinto beans, and cooked spinach all have more than 200 micrograms in a 1-cup serving.

We are all exposed to cancer-causing chemicals despite our

efforts to avoid them. Some people are smokers, and quitting smoking is a vital step for them. But all of us are exposed to chemicals in the air, water, food, and household products in addition to the carcinogens produced within our bodies as a part of our metabolic processes. While trying to minimize our exposure to carcinogens, we can also shore up our defenses against these assaults by including generous amounts of vegetables and fruits in our diet.

Foods and Immunity

Even if we follow healthy lifestyles, cancer cells will arise in the body from time to time. Luckily, we have white blood cells that roam our bloodstreams looking for these troublemakers. Some white blood cells, called natural killer cells, seek out and destroy cancer cells and bacteria. They engulf and destroy aberrant cells before they can cause damage. The function of natural killer cells and other white blood cells is improved by as little as 30 milligrams of beta-carotene per day—the amount in two large carrots.

Although beta-carotene is safe, even in fairly substantial amounts, the best way to get beta-carotene is not in pills, but in carrots, winter squash, spinach, kale, and the other packages in which nature supplies it. Beta-carotene is only one of perhaps two dozen related substances called carotenoids that occur naturally in vegetables and fruits and have varying degrees of biological activity. In a large research review, cancer experts found significant evidence suggesting that carotenoids help protect against breast and lung cancers.⁴

In addition to their antioxidant effects, vitamins C and E and selenium bolster immune function, but the importance of these effects in protecting against cancer is not yet clear.

Too much fat in the diet can impair immunity, and cutting down on fat helps strengthen the immune defenses against cells that turn cancerous. Researchers in New York tested the effect of low-fat diets on immunity.⁶ They put healthy volunteers on a diet that limited fat content to 20%, reducing all fats and oils—not just saturated or unsaturated fats. Three months later, the researchers took blood samples from the volunteers and examined their natural killer cells. The natural killer cell activity was greatly improved.

It should come as no surprise that people who eat plant-based diets have stronger immune systems than meat eaters. Studies of white blood cell samples from vegetarians have shown them to have more than double the cancer-cell-destroying ability of their nonvegetarian counterparts.⁷ The immune-boosting power of plant foods is partly due to their vitamin content, low fat content, and perhaps other contributors, such as reduced exposure to toxic chemicals and animal proteins.

Dietary fat intake can also contribute to weight gain, which can impair immune function. Studies show that individuals who are overweight are at increased risk for various infections and to certain forms of cancer, especially postmenopausal breast cancer.⁸

Foods and Hormones

Several of the most common forms of cancer are linked to sex hormones. This is true of cancers of the breast, uterus, ovary, prostate, and perhaps other sites. The amount of hormones in our bodies and their actions are determined, in large part, by the foods we eat.

Antioxidants in Foods	Vit C (mg)	B-carotene (mg)	Vit E (mg)
Apple (1 medium)	8	0.05	0.38
Broccoli	51	0.73	1.13
Brown rice	0	0	0.34
Brussels sprouts	48	0.36	0.34
Carrot (1 medium)	7	9.11	0.73
Cauliflower	28	0.04	0.04
Chickpeas	2	0.02	0.57
Corn	0	0.16	0.81
Grapefruit (pink, 1/2)	38	0.84	0.16
Navy beans	2	0	0.02
Orange (1 medium)	70	0.09	0.24
Orange juice	84	0.02	0.50
Pineapple	79	0.06	0.03
Soybeans	3	0.01	0.60
Fresh spinach	8	1.69	0.61
Strawberries	89	0.01	0.44
Sweet potato (1 medium with skin)	39	23.02	1.42

Serving size: 1 cup (8 oz.), except as otherwise noted.

Source: US Department of Agriculture, Agricultural Research Service, Nutrient Data Laboratory. USDA National Nutrient Database for Standard Reference, Release 28. Version Current: May 17, 2016. Available at: <https://ndb.nal.usda.gov/ndb/nutrients/index>

Breast Cancer: Prevention

As long ago as 1982, the National Research Council in a report called “Diet, Nutrition, and Cancer,” showed the evidence already available linking specific dietary factors to cancer of the breast and other organs.

International comparisons offer a good illustration. Asian countries, such as Japan, have low rates of breast cancer, while Western countries have cancer rates many times higher. However, when Japanese girls are raised on Westernized diets, their rates of breast cancer increase dramatically.

The traditional Japanese diet is much lower in fat, especially animal fat, than the typical Western diet. In the late 1940s, when breast cancer was particularly rare in that country, less than 10% of the calories in the Japanese diet came from fat.⁹ The American diet is centered on animal products, which tend to be high in fat and low in other important nutrients. The fat content of the average American diet is well above 30% of calories.

It has long been known that countries with a higher intake of fat, especially animal fat, have a higher incidence of breast cancer.¹⁰ Even within Japan, affluent women who eat meat daily have an 8.5 times higher risk of breast cancer than poorer women who rarely or never eat meat.¹¹ In more recent research, it has been found that consuming more healthful plant-based foods, such as whole grains, fruits, and vegetables, is associated with a lower risk of developing breast cancer.^{12,13}

Cutting fat out of the diet helps
strengthen the immune defenses
against cells that turn
cancerous.



Fat and Hormonal Effects

Fatty foods have a strong influence on hormonal activity in the body. First, high-fat diets increase the amount of estrogens, the female sex hormones, in the blood.¹⁴ This is a problem, because it is well known that many breast tumors are “fueled” by estrogens. Estrogens are normal and essential hormones for both women and men, but the more estrogen present, the greater the driving force behind some kinds of breast cancer. On high-fat diets, estrogen levels increase. When women adopt low-fat diets, their estrogen levels drop noticeably in a very short time. In addition, they have more of certain carrier molecules, called sex hormone-binding globulin, that circulate in the blood and have the job of holding onto sex hormones, keeping them inactive until they are needed. Fatty foods do the reverse: They increase estrogens and reduce the amount of the carrier molecule that is supposed to keep estrogens in check.

Animal fats are apparently a bigger problem than vegetable oils. Paolo Toniolo of the New York University Cancer Institute compared the diets of 250 women with breast cancer to 499 women without cancer from the same province in northwestern Italy. The two groups ate about the same amount of olive oil and carbohydrates. What distinguished the cancer patients was that they had eaten more meat, cheese, butter, and milk. Women who consumed more animal products had as much as three times the cancer risk of other women.¹⁵

A 2003 Harvard University study that included more than 90,000 women showed that the women who had eaten the most animal fat had significantly higher risk of breast cancer, compared with women who ate the least.¹⁶

Even though cross-cultural comparisons have pointed a finger at animal fat as the principal problem, vegetable oil is also under some suspicion. A 2009 study conducted by the National Institutes of Health found a direct link with total dietary fat and postmenopausal invasive breast cancer.¹⁷ Vegetable oils can probably affect estrogen levels and, as we will see, can increase the production of cancer-causing free radicals. So it is no good just replacing fried chicken with fried onion rings. The best diet eliminates animal products and keeps vegetable oils to a minimum as well.

Certain foods have special benefits. Soybeans, for example, contain natural compounds called isoflavones. Isoflavones can act to balance the body's natural estrogen by binding to either the alpha- or beta-estrogen receptors. They have particular affinity for estrogen receptor beta, which, in breast tissue, is involved in anti-estrogenic effects. The result is reduced estrogen stimulation of each cell. Research studies have shown that soy foods are not only safe but helpful for both preventing and surviving breast cancer. Eating soy products during adolescence seems to offer the greatest protection for reducing breast cancer risk.¹⁸ Women eating the most soy foods have a lower chance of developing pre- and postmenopausal breast cancer, compared with those who eat the least.¹⁹ Similarly, breast cancer survivors who eat more soy foods have less risk of recurrence and better chance of survival, compared with

women who do not eat as much.²⁰ Soybeans are a mainstay of Asian diets and may be an additional reason why these countries have low cancer rates. A 2012 analysis that combined the results of prior studies, including a total of 9,514 women from the United States and China who had had breast cancer, showed that those who consumed the most soy products were 25% less likely to have their cancer return, compared with those who tended to avoid soy products.²¹

How Much Fat Is Too Much?

The National Cancer Institute has long recommended that fat be limited to less than 30% of calories and that the fattiest meats be replaced by leaner meat, poultry, fish, and vegetables. These recommendations, however, are much too weak to prevent cancer or to increase survival for those already diagnosed with the disease. A large study of American nurses showed that those who limited fat to 27% of their calories were not any better off in preventing cancer than those consuming more fat.²² Some have interpreted this to mean that diet has nothing to do with breast cancer. A more reasonable conclusion is that the diets these women followed were still high-risk diets. After all, a diet including regular consumption of animal products and drawing nearly 30% of calories from fat is nothing like the traditional plant-based Asian diets associated with low cancer risk.

Favoring Fiber

Grains, vegetables, fruits, and legumes ensure plenty of fiber, but chicken breasts, beef, pork, eggs, cheese, and all other animal products contain no fiber at all.

Getting the fat off your plate is important—but that is just the first step. Other food choices play important roles in cancer prevention. Vegetables, fruits, grains, and beans provide fiber, which helps the body dispose of excess estrogen.

One way the body rids itself of sex hormones is through the digestive tract. The liver pulls sex hormones from the blood, chemically alters them, and sends them down the bile ducts into the intestinal tract. There, the fiber from grains, vegetables, fruits, and beans ushers sex hormones through the intestine and out the door as wastes. At least, that is how the system is supposed to work. But chicken, beef, eggs, cheese, and all other animal products contain no fiber at all. As these products have taken up more and more space on the American plate, they have displaced grains, vegetables, beans, and fruits. Without adequate fiber to hold them in the digestive tract, sex hormones are reabsorbed into the bloodstream, where they once again become biologically active. The hormones your body was trying to eliminate are thus put back into circulation.

Building your diet from grains, vegetables, fruits, and legumes ensures plenty of fiber for the body's needs. To increase your daily fiber intake, start your day with oatmeal and fruit, and try adding delicious, low-fat, fiber-rich recipes such as Three-Bean Chili or Easy Bean Salad to your weekly menu (see recipes on page 19).

Carcinogenic Compounds in Cooked Meat

Not only is meat devoid of fiber and other nutrients that have a protective effect, but it also contains animal protein, saturated fat, and, in some cases, carcinogenic compounds formed during the process of cooking meat. These carcinogenic compounds may be to blame for part of the correlation between meat intake and increased cancer risk. Meat intake has been shown to be a risk factor for breast cancer even when researchers controlled for confounding factors such as total fat and calorie intake.²³

Heterocyclic Amines

Heterocyclic amines (HCAs) are DNA-damaging compounds that are produced as meat cooks. Grilling, frying, or oven-broiling meat produces large quantities of these mutagens. The longer and hotter the meat is cooked, the more of these compounds form. In some studies, grilled chicken formed higher concentrations of these cancer-causing substances than other types of cooked meat.²⁴

The major classes of HCAs are formed from creatine or creatinine, specific amino acids, and sugars. All meats, including fish, are high in creatine. HCA formation is greatest when meat is cooked at high temperatures, as is most common with grilling or frying. Consumption of well-done meat has been associated with increased risk of breast cancer²⁵ and colon cancer.²⁶

Polycyclic Aromatic Hydrocarbons

Grilling or broiling meat over a direct flame results in fat dropping on the hot fire and the production of flames containing polycyclic aromatic hydrocarbon (PAH). PAHs adhere to the surface of food; the more intense the heat, the more PAHs are present.⁴ They are believed to play a significant role in contributing to human cancers.

Other Factors

Aside from diet, other factors that increase breast cancer risk include the following:

Hormones: Supplemental hormones given to women after menopause (average age 63.3) were found to slightly increase breast cancer risk, although the findings were not statistically significant.²⁷ Oral contraceptives may also increase risk slightly. Although newer birth control pills contain less estrogen and progesterone than older versions, evidence suggests some increase in risk.²⁸ After stopping birth control for at least five years, risk returns to about equal to those who have never used birth control.²⁹

Overweight: Higher body weight increases the risk of breast cancer after menopause.³

Radiation: Of all the different parts of the body, the breast is one of the most sensitive to radiation, which may be a risk factor in the development of breast cancer. In order to reduce the risk of breast cancer from radiation exposure, such as X-rays and mammograms, exposure should only be applied

if the benefit outweighs the risk.³⁰ Annual mammograms are clearly beneficial for women over 50. But women should schedule mammograms only at modern facilities that do them regularly and maintain new equipment, which keeps radiation doses to a minimum. For women under 50, it is unclear whether routine mammograms are beneficial. Many cancers are missed on mammograms, and women have sometimes been falsely reassured by a negative mammogram, leading to delays in diagnosis and treatment. All women (especially those under 50) should do regular self-exams and follow their physicians' recommendations regarding mammogram screening.

Genetics: About 5% of breast cancer cases are purely attributable to genetics. In such cases, cancer is passed from parent to child as a dominant trait, and the family tree is riddled with the disease. For a larger group of individuals, genetics probably makes a contribution in subtle ways. For example, it may well be that different genes influence one's susceptibility to carcinogens, the strength of the immune system, body weight, and other factors. Each of these is also influenced by diet.

Toxic Chemicals: Populations in areas near toxic waste sites tend to have higher than average rates of breast cancer. That is true for other forms of cancer, too. And you don't have to live near a chemical waste site to be concerned about toxic exposures. Toxic chemicals are available at any grocery store. Chemical contaminants frequently end up in meats and dairy products, because pesticides are sprayed on grains fed to cows, chickens, pigs, and other livestock. In storage bins, feed grains are sprayed again. Animals concentrate these chemicals in their tissues. Meat, fish, and milk are the largest sources.^{31,32} Traces of pesticides can also be found on nonorganic produce. Fortunately, organic produce is now more widely available.

Women who avoid eating animal products have much lower concentrations of pesticides in their breast milk. It appears eggs, milk, meat, fish, and shrimp can contribute to higher levels of pesticides in pregnant women.³³ Pesticides such as DDT, chlordane, heptachlor, and dieldrin, and polychlorinated biphenyls (PCBs) have been measured at markedly lower levels in vegetarians than in omnivores. In a 1981 study, vegetarians had only 1-2% of the national average levels of certain pesticides and industrial chemicals compared with levels in average Americans.³⁴ The exception was PCBs, for which vegetarians had levels comparable to meat-eaters. PCBs in the body often reflect past fish consumption, and levels drop slowly after people adopt a plant-based diet. Once PCBs are in the body tissues, avoiding contaminated fish will reduce PCB levels only very slowly.

Time Between Puberty and First Pregnancy: The younger a girl is when puberty occurs, the higher her risk of breast cancer. Also, the later the age of her first pregnancy, the higher her risk. However, it may be that the early age of puberty simply indicates elevated hormone levels, as described above. As high-fat, low-fiber diets have spread from the wealthy to most of the population, the age of puberty has dropped dramatically from 17 in 1840 to 12.5

today. Similarly, as Japan's diet has Westernized since World War II, the age of puberty has dropped from 15 to 12.5. It may be that early puberty and cancer are both the result of a hormonal aberration.

Cancers of the Uterus and Ovary: Prevention

The uterus and ovary are reproductive organs, and factors that affect hormone function can be expected to affect these organs as well. The risk of cancer of the uterus and ovary is higher in populations that have more breast cancer incidences, suggesting that they may be caused by similar factors. Obesity after menopause is also a strong yet modifiable risk factor for epithelial ovarian cancer risk.³⁵ Uterine cancer is linked to fatty diets and obesity, although other factors, including hormone supplements, also play an important role.³⁶

The consumption of dairy products may also contribute to development of ovarian cancer. The relationship between dairy products and ovarian cancer may be due to the breakdown of the milk sugar (lactose) into galactose, a sugar that may be toxic to ovarian cells.³⁷ The problem is the milk sugar, not the milk fat, so it is not solved by using nonfat products. Cruciferous vegetables, such as broccoli, cabbage, and cauliflower, in particular, have been found to reduce the risk of developing ovarian cancer and to improve survival in women diagnosed with ovarian cancer. It has been suggested that the isothiocyanates found in cruciferous vegetables have anticancer properties.³⁸

Prostate Cancer: Prevention

Like women, men on high-fat Western diets have more estrogens circulating in their blood and a higher risk of cancer of reproductive organs. High-fat diets alter the amounts of testosterone, estrogen, and other hormones in both men and women. Cancer of the prostate, which occurs primarily in older individuals, is the most common form of cancer in American men.

Cancer cells are found in the prostates of about 30% of men over the age of 50.³⁹ In most cases, these cancer cells do not develop into cancerous tumors that affect the overall health or life span of the individual. However, in some cases, the cancer does grow, invade surrounding tissues, and spread to other parts of the body. Although the disease varies greatly from one person to the next, the average patient loses nine years from his normal life span. One in 10 men will develop prostate cancer at some point in his life.

Asian and Latin American countries have a much lower prevalence of prostate cancer, but the disease is very common in Europe and North America. Ten men die of prostate cancer in Western Europe for every one who dies of the disease in Asia.

Cancer of the prostate is strongly linked to what men eat. Milk, meat, eggs, cheese, cream, butter, and fats are found to be linked to prostate cancer.⁴⁰ A study looking at nearly 2 million men and more than 35,000 new prostate cancer diagnoses found that consuming processed meat was linked to an increased risk of both total and advanced prostate cancer.⁴¹

Milk consumption has been linked to prostate cancer due to

high levels of the compound insulin-like growth factor (IGF-1), which is found both in dairy products and in increased levels in the bodies of those who consume dairy on a regular basis.⁴²

A Harvard University study showed that men who had the highest levels of IGF-1 had more than four times the risk of prostate cancer, compared with those who had the lowest levels.⁴³

In addition, two more Harvard studies have shown that milk-drinking men have 30-60% greater prostate cancer risk than men who generally avoid dairy products.^{44,45}

One glass of skim milk contains 8.4 grams of protein. Researchers estimate that every 35 grams of dairy protein consumed daily can increase prostate cancer risk by 32%.⁴⁶ Calcium from dairy products was also positively associated with risk, supporting the hypothesis that high intake of protein or calcium from dairy products may increase prostate cancer risk.

Who has a lower risk? Countries with more rice, soybean products, or green or yellow vegetables in the diet have far fewer prostate cancer deaths. Diets rich in lycopene, the bright red pigment found in tomatoes, watermelon, and pink grapefruit, have also been shown to prevent prostate cancer. Researchers found that men who had just two servings of tomato sauce per week had a 23% lower risk of prostate cancer, compared with those who rarely ate tomato products.⁴⁷ In another study, researchers found that those who followed a vegan diet experienced a 35% lower prostate cancer risk than those following a nonvegetarian, lacto-ovo-vegetarian, pesco-vegetarian, or semi-vegetarian diet.⁴⁸

Men who consume diets based on animal products tend to have higher levels of testosterone compared with men who eat plant-based diets. This increase may be due to overproduction of these hormones in the body. Also, fiber in the diet helps remove excess hormones with body wastes. Those who eat meat and dairy products miss out on a substantial amount of fiber, because animal products have no fiber at all. This hormonal boost can affect the prostate, which is the likely reason for increased cancer risk among men on meat-based diets.

Colon Cancer: Prevention

The colon is another name for the large intestine, which makes up the second half of our digestive tract. Strong links



have been found between colon cancer and the consumption of alcohol, meat, and other fatty foods.⁴

Red and processed meat products (bacon, ham, lunch meat, sausage) are the biggest offenders. A substantial body of scientific evidence attributes the consumption of processed meat to significantly increased cancer risk. Critically, a 2015 review by the World Health Organization provided further evidence that processed meat consumption is “carcinogenic to humans.”⁴⁹

The World Cancer Research Fund, in conjunction with the American Institute for Cancer Research, found, in a comprehensive and ongoing analysis of research, that red and processed meat are, together and individually, “convincing” as risk factors for colorectal cancer. The combination of evidence indicates a 30-50% increased risk for colorectal cancer when consumption is highest.⁵⁰

Researchers also established the dose-response nature of red and processed meat consumption and colorectal cancer risk. Consuming 100 grams of red meat per day, or a steak the size of the palm of your hand, has been found to increase the relative risk for colorectal cancer by 10% and colon cancer by 17%.⁵¹

There are several explanations for the causal association between red and processed meat and cancer, including DNA-damaging n-nitroso compounds, heme iron, heterocyclic amines, and polycyclic aromatic hydrocarbons, among others.^{52,53} Just as tobacco causes cancer by more than one presumed mechanism, the same is likely true for red and processed meat. Foods high in saturated fat and cholesterol are also associated with greater colorectal cancer risk.⁵⁴⁻⁵⁶

High-fiber diets offer a measure of protection. Fiber greatly speeds the passage of food through the colon, effectively removing carcinogens. And fiber actually changes the type of bacteria present in the intestine, which reduces production of carcinogenic secondary bile acids. Fiber also absorbs and dilutes bile acids. A meta-analysis found a 12% lower risk for colorectal cancer in those who ate the most total dietary fiber.⁵⁷

Even people at particular risk for cancer can be helped by

high-fiber diets. Jerome J. DeCosse, MD, a surgeon at Cornell Medical Center, gave bran to patients with recurrent polyps of the colon. These are small growths that have a tendency to become cancerous. Dr. DeCosse found that, within six months, the polyps became smaller and fewer in number. He believes that pentose fiber, which is plentiful in wheat, is the key to bran’s power.⁵⁸

A 2011 study published in the *British Medical Journal* found that high-fiber foods, whole grains in particular, significantly reduced the risk of colon cancer.⁵⁹ People eating three servings of whole grains each day could decrease their risk by 17%.⁶⁰

Obesity also influences cancer risk. There is convincing evidence that abdominal and body fat increases one’s risk of colon cancer.⁴ Avoiding obesity is one of the greatest preventative steps for significantly lowering cancer risk, as a higher body mass index, or BMI, is a risk for colon cancer.⁶¹ The good news: Fiber is important for helping maintain a healthy body weight.

Plant Foods: Powerful for Health

Two themes consistently emerge from cancer research: Vegetables and fruits help reduce risk, while animal products and other fatty foods are frequently found to increase risk.

When the terms “fiber” and “fat” are used, it is easy to forget the foods from which they come. When you hear about the dangers of fat, think meat- and- dairy-based diets, aided and abetted by oily foods. Fiber is found in whole grains, vegetables, fruits, and beans. There is no fiber in any product from an animal.

A vegan menu is a powerful and pleasurable way to achieve good health. The vegan eating pattern is based on a wide variety of foods that are satisfying, delicious, and healthful. Are you ready to get started?



Even people at particular risk
for cancer can be helped by
high-fiber diets.

Try the Power Plate and Discover a Healthier Way To Live!



Vegetables

Vegetables are packed with nutrients, including vitamin C, beta-carotene, riboflavin, iron, calcium, fiber, and other nutrients. Dark-green leafy vegetables, such as broccoli, collards, kale, mustard and turnip greens, chicory, and bok choy, are especially good sources of these important nutrients. Dark-yellow and orange vegetables such as carrots, winter squash, sweet potatoes, and pumpkin provide extra beta-carotene. Include generous portions of a variety of vegetables in your diet.

Whole Grains

This group includes bread, rice, pasta, hot or cold cereal, corn, millet, barley, bulgur, buckwheat groats, and tortillas. Build each of your meals around a hearty grain dish. Grains are rich in fiber and other complex carbohydrates, as well as protein, B vitamins, and zinc.

Fruit

Fruits are rich in fiber, vitamin C, and beta-carotene. Be sure to include at least one serving each day of fruits high in vitamin C—citrus fruits, melons, and strawberries are all good choices. Choose whole fruit over fruit juices, which do not contain very much fiber.

Legumes

Legumes, including beans, peas, and lentils, are all good sources of fiber, protein, iron, calcium, zinc, and B vitamins. This group also includes chickpeas, baked and refried beans, soy milk, tempeh, and texturized vegetable protein.

Be sure to include a good source of vitamin B12, such as fortified cereals or vitamin supplements.

Setting Blame Aside

Occasionally people who have cancer report feeling that, if food plays a role in cancer, then they are somehow to blame for their disease. As such, guilt and blame often become concerns for people dealing with cancer. However, these feelings are burdens that help no one. Besides, it makes no sense to blame anyone for things they had no way of knowing. Until major public education programs spread the word about the role of dietary factors and help people to change, cancer will remain an epidemic.

Steps to Cancer Prevention⁴

- Maintain your weight at or near your ideal level.
- Be physically active.
- Eat a diet rich in whole grains, vegetables, fruits, and legumes.
- Limit consumption of fast foods and other processed foods high in fat and sugar.
- Drink water and unsweetened drinks.
- Minimize alcohol intake.
- Be sure to take a daily multivitamin without added minerals, or other reliable source of vitamin B12.

Surviving Cancer

Foods are important, not only in preventing cancer, but also in improving survival for those who already have cancer.

Breast Cancer: Survival

Not all cancers of the breast are the same. Some have a relatively good prognosis, and others have a very poor prognosis. For example, a tumor that is small and has not spread to the lymph nodes or other organs is less dangerous than a tumor that is larger and has already spread. (Lymph nodes are pea-sized collections of cells near the breast and other organs; they are important to immune function.) Hospital laboratories also determine whether a breast tumor has receptors for estrogen or progesterone hormones. If it does, the tumor is slightly less aggressive than a tumor lacking receptors.

These prognostic factors are not due to chance alone. Decades ago, Ernst Wynder of the American Health Foundation in New York observed that Japanese women are much less likely than American women to get breast cancer; in addition, when Japanese women do get the disease, they tend to survive longer.⁶² Their improved survival is independent of age, tumor size, estrogen receptor status, the extent of spread to lymph nodes, and the microscopic appearance of the cancer cells. And it is not that Japanese women have better health care, because the same pattern has been observed in Hawaii and California, where Japanese women live near other ethnic groups and have essentially the same health care system.

Researchers have begun to look at whether diet plays a role in survival. It does. Our old enemy, fat in foods, rears its ugly head once again—the more fat in the diet, the shorter the survival time. In one Canadian research study, women with cancer were more likely to have lymph node involvement if they had a higher fat intake. This effect was found only for saturated fat and only for postmenopausal women.⁶³ Fat seems to have a measurable effect when cancer has spread to other parts of the body and little or no effect when the disease is localized.

Researchers in Buffalo, N.Y., calculated what they believe to be the degree of risk posed by fat in the diet: For a woman with metastatic breast cancer (cancer that has already spread at the time of diagnosis), the risk of dying from the disease at any point increases 40% for every 1,000 grams of fat consumed monthly.⁶⁴ To understand what this means, compare three different diets, each of which contains 1,800 calories per day:

- On a low-fat plant-based diet, about 10% of calories come from fat. This type of diet contributes about 20 grams of fat per day, or 600 grams per month.
- On a typical American diet, 35% of calories come from fat. This means about 70 grams of fat per day, or 2,100 grams per month.
- On a diet with more fat than average, say 50% of calories, fat intake would be 100 grams per day, or 3,000 grams per month.

If the researchers' finding holds, the typical American diet would lead to about a 60% higher risk of dying of breast cancer at any given point, compared with the low-fat vegan diet, and the high-fat diet would lead to a more than 95% increase in risk of dying. These figures do not mean that a woman's risk of dying is 60% or 95%. They mean that the risk is 60% or 95% higher than it would otherwise have been, assuming the individual is comparable to those studied.

In overweight women with metastatic breast cancer, a whole food, plant-based diet can improve health markers. After just eight weeks on the diet, women in the intervention group of this research study had reduced fasting insulin, reduced insulin resistance, reduced total and LDL cholesterol, an improved hormonal profile, and an average weight loss of 6.6%.⁶⁵

Other parts of the diet play important roles. Diets high in fiber, carbohydrates, and vitamin A seem to help the prognosis, while alcohol slightly worsens it.⁶⁶ Patients who have more estrogen receptors on their tumors—which indicates a better prognosis—tend to be those who had consumed more vitamin A. (Beta-carotene becomes vitamin A in the body.) For reasons that are not entirely clear, vegetables and fruits (and the vitamins they contain) help keep the cells of the body in better working order—one sign of which, for breast cells, is the presence of estrogen receptors. So vegetables and fruits are not only important in helping to prevent cancer, they also help improve survival for those who have cancer.

Higher body weight increases the risk of dying of breast cancer.⁶⁷ Among postmenopausal women with breast cancer,

slimmer women tend to have less lymph node involvement. Heavier women have more lymph node involvement, higher rates of recurrence, and poorer survival rates. Even among relatively slim women, it pays to avoid extra weight. A 2006 study from Shanghai, China, showed that women who had previously been diagnosed with breast cancer had greater survival rates if they were at the slimmer part of the normal weight range (body mass index below 23), compared with even slightly heavier women.⁶⁸

Cancers of the Uterus and Ovary: Survival

The uterus and ovary, like the breast, are strongly influenced by sex hormones. Again, a low-fat vegan diet is the best prescription for preventing the hormonal elevations that encourage cancer. In addition, as mentioned earlier, galactose—a product of the breakdown of the milk sugar lactose—may increase the risk of ovarian cancer.^{27,28}

A meta-analysis published in 2020 found that women who ate the most fruit and vegetables before a diagnosis of ovarian cancer had an 18% and 22% lower risk of dying, respectively.⁶⁹

It's a good idea for those with ovarian or uterine cancer to follow the same diet that helps prevent cancer in these organs and that keeps the immune system in good working order: a low-fat vegan diet rich in vegetables and fruits.

Maintaining a healthy weight and following a low-fat, plant-based diet is the best prescription for preventing the hormonal elevations that encourage cancer.^{69,70}

Prostate Cancer: Survival

Diet may help improve survival in prostate cancer as well. When pathologists conduct autopsies of men who die from accidents or other causes, they find cancer cells in the prostates of about 20%. These men did not know they had cancer and had no symptoms whatsoever. The prevalence of such latent cancers actually varies with location: The lowest rates are in Singapore (13%) and Hong Kong (15%), and the highest are in Sweden (31%).⁷¹ In most men, the cells never grow into a large tumor, never spread, and never affect life or health in any way. However, just as the prevalence of latent cancers varies from one country to another, the likelihood that they will turn into symptomatic cancer varies in precisely the same way, suggesting that the same factors that cause cancer cells to form in the first place also encourage them to grow and spread. So while a Swede is twice as likely as a man from Hong Kong to have cancerous cells in his prostate, he is more than eight times more likely to die of prostate cancer.

A low-fat, high-fiber diet can help eliminate the hormonal aberrations that are known to be linked with prostate cancer and may help improve survival among those who have the disease.⁷²

In a study of men with prostate cancer, Dean Ornish, MD, tested a low-fat vegan diet accompanied by regular exercise and stress management. In the 42 men in the control group, the amount of prostate-specific antigen (PSA, which is used as an indicator of cancer advancement) levels rose over the

three-month study period, and seven required additional treatment. But in the 42 men assigned to the vegan diet and lifestyle intervention, the average PSA level dropped from 6.3 to 5.8, and none required further treatment.⁷³ Another study of more than 2,000 men with prostate cancer found that those who ate the most plant-based foods had a 52% and 53% lower risk of progression and recurrence, respectively.⁷⁴

Colon Cancer: Survival

Colon cancer is encouraged by diets containing animal fat and discouraged by diets rich in vegetables and fiber. A low-fat plant-based diet is important both for those seeking to prevent cancer and those who have already been treated for it. In a prospective study evaluating 1,009 colon cancer patients, researchers found that survival depended to a great extent on dietary habits. The participants had completed initial cancer treatment. Those who consumed more red and processed meats, sweets, and refined grains were more likely to have a recurrence or die from the disease after a median 5.3-year follow-up, compared with those who consumed more fruits, vegetables, whole grains, and less red and processed meats and refined foods.⁷⁵

Researchers at the University of Arizona found that people who have been treated for colon or rectal cancer have less risk of recurrence when their diets are rich in fiber. They found benefits from daily supplements of 13.5 grams of wheat bran fiber (the amount in a half-cup of bran cereal), but they speculate that other forms of fiber might have the same effect.⁷⁶ A vegan diet can easily boost fiber intake by 10 to 29 grams per day. If you have bran cereal, topping it with soy milk rather than cow's milk allows you to avoid animal fat, cholesterol, lactose, and animal proteins.

Colon cancer typically develops from polyps in the colon wall. These polyps become smaller and fewer in number within six months on a high-fiber diet.

It is clear that much more needs to be learned about the power of foods to prevent cancer or to improve cancer survival. The good news is that the diet that helps protect against cancer is the same one that keeps cholesterol low and waistlines slim. Eliminating animal products from the diet, keeping oils to a minimum, and including generous amounts of vegetables, grains, beans, and fruits is a powerful prescription.

Making Healthy Diet Changes

The Three-Step Way to Go Vegan

If you are making the switch to a vegan diet for its health benefits, you'll be pleased to discover a wonderful additional benefit to vegan eating: It's a delicious and fun way to explore new foods. A vegan meal can be as familiar as spaghetti with marinara sauce, as comforting as a bowl of rich potato soup, or as refreshing as Cucumber, Mango, and Spinach Salad (see the recipe on page 17).



The switch to a vegan diet is easier than you might think. Most people, whether vegan or meat eaters, typically use a limited variety of recipes; the average family eats only eight or nine different dinners repeatedly. You can use a simple, three-step method to come up with nine vegan dinner menus that you enjoy and can prepare easily.

First, think of three vegan meals you already enjoy. Common ones are pasta primavera, vegetable stir-fry, and vegetable stew.

Second, think of three recipes you prepare regularly that can easily be adapted to a vegan menu. For example, a favorite chili recipe can be made with almost the same ingredients; just replace the meat with beans or texturized vegetable protein. Prepare bean burritos (using canned vegetarian refried beans) instead of beef burritos, veggie burgers instead of hamburgers, and grilled eggplant and roasted red peppers instead of grilled chicken in sandwiches. Many soups, stews, and casseroles also can be made into vegan dishes with a few simple changes.

Third, check out some vegan cookbooks from the library and experiment with the recipes for a week or so until you find three new recipes that are delicious and easy to make. You can also explore the many recipes offered on our website, PCRM.org/Recipes. Just like that, with minimal changes to your menus, you will have nine vegan dinners.

After that, coming up with vegan options for breakfast and lunch is easy. Try muffins with fruit spread, cholesterol-free French toast, or cereal for breakfasts. Sandwiches, with spreads like hummus or white bean pâté with lemon and garlic, pasta salads, or even dinner leftovers make great lunches.

Tips for Making the Switch to a Vegan Diet

- Convenience foods cut cooking time. Supermarkets and natural foods stores stock a huge array of instant soups and main-dish vegan convenience items. Many canned soups, such as minestrone, black bean, or vegetable, are vegan. Flavored rice or other grain mixes, like curried rice or tabbouleh salad, can be stretched into an entree with a can of beans. Visit the frozen food section for internationally inspired plant-based

frozen entrees such as corn and bean enchiladas, lentil curry, or vegan pad thai. Or try vegetarian baked beans, refried beans, sloppy joe sauce, and meatless spaghetti sauce from the canned goods aisle.

- Ask for it! Even restaurants that don't offer vegan entrees can usually whip up a meatless pasta or vegetable plate if you ask. If attending a catered affair, catch the waiter before you are served and ask them to remove the chicken breast from your plate and slip on an extra baked potato.
- When you're dining out, the best bets for finding vegan food are international restaurants. Italian, Chinese, Mexican, Thai, Japanese, and Indian restaurants all offer a wide variety of plant-based dishes.
- Summer barbecues are healthy and fun with meatless hot dogs and burgers, which are now available in most supermarkets. Or for a real change of pace, grill thick slices of marinated vegetables like eggplant, zucchini, or tomatoes.

Top Five Foods You Can't Live Without

1. Beans and lentils add heartiness to soups, stews, chili, and other recipes. Plus, they're loaded with cancer-fighting fiber and healthy plant protein.
2. Broccoli and other cruciferous vegetables such as kale, cauliflower, collard greens, and Brussels sprouts contain potent cancer-fighting compounds and have been shown to help rid the body of excess hormones.
3. Berries such as blueberries and blackberries are full of powerful antioxidants that help stop free radical damage which can otherwise lead to cancer.
4. Carrots and other orange-colored fruits and vegetables such as sweet potatoes, mango, cantaloupe, pumpkin, and squash are incredible sources of beta-carotene, which is crucial for cancer prevention and survival. Just one large carrot a day helps you meet your daily requirement!
5. Whole grains such as brown rice, quinoa, oats, and millet are rich sources of fiber and B vitamins, essential for health. Plus, they're loaded with complex carbohydrates to keep you full and help you avoid unhealthy snacking.



References

- 2024 Cancer Facts & Figures Cancer | 2024--First Year the US Expects More than 2M New Cases of Cancer. www.cancer.org/research/acs-research-news/facts-and-figures-2024.html#:~:text=In%202024%2C%20over%2061%2C000%20deaths
- Islami F, Marlow EC, Thomson B, et al. Proportion and number of cancer cases and deaths attributable to potentially modifiable risk factors in the United States, 2019. *CA Cancer J Clin*. 2024;74(5):405-432. doi:10.3322/caac.21858
- Minamoto T, Mai M, Ronai Z. Environmental factors as regulators and effectors of multistep carcinogenesis. *Carcinogenesis*. 1999;20:519-527.
- World Cancer Research Fund / American Institute for Cancer Research. *Food, Nutrition, Physical Activity, and the Prevention of Cancer: a Global Perspective*. Washington DC: AICR, 2018.
- García-Closas R, Castellsague X, Bosch X, Gonzalez CA. The role of diet and nutrition in cervical carcinogenesis: a review of recent evidence. *Int J Cancer*. 2005;117:629-637.
- Barone J, Hebert JR, Reddy MM. Dietary fat and natural killer cell activity. *Am J Clin Nutr*. 1989;50:861-867.
- Malter M, Schriever G, Eilber U. Natural killer cells, vitamins, and other brood components of vegetarian and omnivorous men. *Nutr Cancer*. 1989;12:271-278.
- Lamas O, Marti A, Martinez JA. Obesity and immunocompetence. *Eur J Clin Nutr*. 2002;56(Suppl 3):S42-S45.
- Lands WEM, Hamazaki T, Yamazaki K, et al. Changing dietary patterns. *Am J Clin Nutr*. 1990;51:991-993.
- Rose DP, Boyar AP, Wynder EL. International comparisons of mortality rates for cancer of the breast, ovary, prostate, and colon, and per capita food consumption. *Cancer*. 1986;58:2363-2371.
- Hirayama T. Epidemiology of breast cancer with special reference to the role of diet. *Prev Med*. 1978;7:173-195.
- Shah S, Mahamat-Saleh Y, Ait-Hadad W, et al. Long-term adherence to healthful and unhealthful plant-based diets and breast cancer risk overall and by hormone receptor and histologic subtypes among postmenopausal females. *Am J Clin Nutr*. 2023;117(3):467-476. doi:10.1016/j.ajcnut.2022.11.019
- Farvid MS, Barnett JB, Spence ND. Fruit and vegetable consumption and incident breast cancer: a systematic review and meta-analysis of prospective studies. *Br J Cancer*. 2021;125(2):284-298. doi:10.1038/s41416-021-01373-2
- Goldin BR, Gorbach SL. Effect of diet on the plasma levels, metabolism, and excretion of estrogens. *Am J Clin Nutr*. 1988;48:787-790.
- Toniolo P, Riboli E, Protta F, Charrel M, Cappa AP. Calorie providing nutrients and risk of breast cancer. *J Natl Cancer Inst*. 1989;81:278-286.
- Cho E, Spiegelman D, Hunter DJ, et al. Premenopausal fat intake and risk of breast cancer. *J Natl Cancer Inst*. 2003;95:1079-1085.
- Thiébaud AC, Kipnis V, Chang SC, et al. Dietary fat and postmenopausal invasive breast cancer in the National Institutes of Health-AARP Diet and Health Study cohort. *J Natl Cancer Inst*. 2007;99:451-462.
- Lee SA, Shu XO, Li H, et al. Adolescent and adult soy food intake and breast cancer risk: results from the Shanghai Women's Health Study. *Am J Clin Nutr*. 2009;89:1920-1926.
- Cho YA, Kim J, Park KS, et al. Effect of dietary soy intake on breast cancer risk according to menopause and hormone receptor status. *Eur J Clin Nutr*. 2010;64:924-932.
- Shu X, Zheng Y, Cai H, et al. Soy food intake and breast cancer survival. *JAMA*. 2009;302:2437-2443.
- Nechuta SJ, Caan BJ, Chen WY, et al. Soy food intake after diagnosis of breast cancer and survival: an in-depth analysis of combined evidence from cohort studies of US and Chinese women. *Am J Clin Nutr*. 2012;96:123-132.
- Willett WC, Stampfer MJ, Colditz GA, et al. Dietary fat and risk of breast cancer. *N Engl J Med*. 1987;316:22-28.
- De Stefani E, Ronco A, Mendilaharsu M, Guidobono M, Deneo-Pellegrini H. Meat intake, heterocyclic amines, and risk of breast cancer: case-control study in Uruguay. *Cancer Epidemiol Biomarkers Prev*. 1997;6:573-581.
- Sinha R, Rothman N, Brown ED, et al. High concentrations of the carcinogen 2-amino-1-methyl-6-phenylimidazo-[4,5] pyridine [PhIP] occur in chicken but are dependent on the cooking method. *Cancer Res*. 1995;55:4516-4519.
- Sinha R, Gustafson DR, Kulldorff M, Wen WQ, Cerhan JR, Zheng W. 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine, a carcinogen in high-temperature-cooked meat, and breast cancer risk. *J Natl Cancer Inst*. 2000;92:1352-1354.
- Butler LM, Sinha R, Millikan RC, et al. Heterocyclic amines, meat intake, and association with colon cancer in a population-based study. *Am J Epidemiol*. 2003;157:434-445.
- Chlebowski RT, Anderson GL, Aragaki AK, et al. Association of menopausal hormone therapy with breast cancer incidence and mortality during long-term follow-up of the Women's Health Initiative randomized clinical trials. *JAMA*. 2020;324(4):369-380. doi:10.1001/jama.2020.9482
- Mørch LS, Skovlund CW, Hannaford PC, Iversen L, Fielding S, Lidegaard Ø. Contemporary hormonal contraception and the risk of breast cancer. *N Engl J Med*. 2017;377(23):2228-2239. doi:10.1056/NEJMoa1700732
- Burchard NA, Eliassen AH, Shafir AL, et al. Oral contraceptive use by formulation and breast cancer risk by subtype in the Nurses' Health Study II: a prospective cohort study. *Am J Obstet Gynecol*. 2022;226(6):821.e1-821.e26. doi:10.1016/j.ajog.2021.12.022
- Golubicic I, Borojevic N, Pavlovic T. Risk factors for breast cancer: is ionizing radiation among them? *J BUON*. 2008;13(4):487-494.
- Schade G, Heinzow B. Organochlorine pesticides and polychlorinated biphenyls in human milk of mothers living in northern Germany: current extent of contamination, time trend from 1986 to 1997 and factors that influence the levels of contamination. *Sci Total Environ*. 1998;215:31-39.
- Gasull M, Bosch de Basea M, Puigdomènech E, Pumarega J, Porta M. Empirical analyses of the influence of diet on human concentrations of persistent organic pollutants: a systematic review of all studies conducted in Spain. *Environ Int*. 2011;37:1226-1235.
- Cao LL, Yan CH, Yu XD, et al. Relationship between serum concentrations of polychlorinated biphenyls and organochlorine pesticides and dietary habits of pregnant women in Shanghai. *Sci Total Environ*. 2011;409:2997-3002.
- Hergenrath J, Hlady G, Wallace B, Savage E. Pollutants in breast milk of vegetarians. *Lancet*. 1981;304:792.
- Lahmann PH, Cust AE, Friedenreich CM, et al. Anthropometric measures and epithelial ovarian cancer risk in the European Prospective Investigation into Cancer and Nutrition. *Int J Cancer*. 2010;126:2404-2415.
- Engeland A, Tretli S, Bjorge T. Height, body mass index, and ovarian cancer: a follow-up of 1.1 million Norwegian women. *J Natl Cancer Inst*. 2003;95:1244-1248.
- Cramer DW, Greenberg ER, Titus-Ernstoff L, et al. A case-control study of galactose consumption and metabolism in relation to ovarian cancer. *Cancer Epidemiol Biomarkers Prev*. 2000;9:95-101.
- Wei YF, Hao YY, Gao S, et al. Pre-diagnosis cruciferous vegetables and isothiocyanates intake and ovarian cancer survival: a prospective cohort study. *Front Nutr*. 2021;8:778031. doi:10.3389/fnut.2021.778031
- Stangelberger A, Walder M, Djavan B. Prostate cancer in elderly men. *Rev Urol*. 2008;10:111-119.
- Park Y, Mitrou PN, Kipnis V, Hollenbeck A, Schatzkin A, Leitzmann MF. Calcium, dairy foods, and risk of incident and fatal prostate cancer: the NIH-AARP Diet and Health Study. *Am J Epidemiol*. 2007;166:1270-1279.
- Nouri-Majd S, Salari-Moghaddam A, Aminianfar A, Larijani B, Esmailzadeh A. Association between red and processed meat consumption and risk of prostate cancer: a systematic review and meta-analysis. *Front Nutr*. 2022;9:801722. doi:10.3389/fnut.2022.801722
- Qin LQ, He K, Xu JY. Milk consumption and circulating insulin-like growth factor-I level: a systematic literature review. *Int J Food Sci Nutr*. 2009;60(Suppl 7):330-340.
- Chan JM, Stampfer MJ, Giovannucci E, et al. Plasma insulin-like growth factor-I and prostate cancer risk: a prospective study. *Science*. 1998;279:563-565.
- Giovannucci E, Rimm EB, Wolk A, et al. Calcium and fructose intake in relation to risk of prostate cancer. *Cancer Res*. 1998;58:442-447.
- Chan JM, Stampfer MJ, Ma J, Gann PH, Gaziano JM, Giovannucci E. Dairy products, calcium, and prostate cancer risk in the Physicians' Health Study. *Am J Clin Nutr*. 2001;74:549-554.
- Allen NE, Key TJ, Appleby PN, et al. Animal foods, protein, calcium and prostate cancer risk: the European Prospective Investigation into Cancer and Nutrition. *Br J Cancer*. 2008;98:1574-1581.
- Giovannucci E, Rimm EB, Liu Y, Stampfer MJ, Willett WC. A prospective study of tomato products, lycopene, and prostate cancer risk. *J Natl Cancer Inst*. 2002;94:391-398.
- Tantamango-Bartley T, Knutsen SF, Knutsen R, et al. Are strict vegetarians protected against prostate cancer? *Am J Clin Nutr*. 2016;103:153-160.
- Bouvard V, Loomis D, Guyton KZ, et al. Carcinogenicity of consumption of red and processed meat. *Lancet Oncol*. 2015;16:1599-1600.
- Vargas AJ, Thompson PA. Diet and nutrient factors in colorectal cancer risk. *Nutr Clin Pract*. 2012;27:613-623.
- Farvid MS, Sidahmed E, Spence ND, Mante Angua K, Rosner BA, Barnett JB. Consumption of red meat and processed meat and cancer incidence: a systematic review and meta-analysis of prospective studies. *Eur J Epidemiol*. 2021;36(9):937-951. doi:10.1007/s10654-021-00741-9
- Loh YH, Jakszyn P, Luben RN, Mulligan AA, Mitrou PN, Khaw KT. N-Nitroso compounds and cancer incidence: the European Prospective Investigation into Cancer and Nutrition (EPIC)-Norfolk Study. *Am J Clin Nutr*. 2011;93:1053-1061.
- Nöthlings U, Wilkens LR, Murphy SP, Hankin JH, Henderson BE, Kolonel LN. Meat and fat intake as risk factors for pancreatic cancer: the multiethnic cohort study. *J Natl Cancer Inst*. 2005;97:1458-1465.



References

54. Mei J, Qian M, Hou Y, et al. Association of saturated fatty acids with cancer risk: a systematic review and meta-analysis. *Lipids Health Dis.* 2024;23:32. doi:10.1186/s12944-024-02025-z
55. Yang Z, Tang H, Lu S, Sun X, Rao B. Relationship between serum lipid level and colorectal cancer: a systemic review and meta-analysis. *BMJ Open.* 2022;12(6):e052373. doi:10.1136/bmjopen-2021-052373
56. Bastide NM, Pierre FH, Corpet DE. Heme iron from meat and risk of colorectal cancer: a meta-analysis and a review of the mechanisms involved. *Cancer Prev Res (Phila).* 2011;4(2):177-184. doi:10.1158/1940-6207.CAPR-10-0113
57. Celiberto F, Aloisio A, Girardi B, et al. Fibres and colorectal cancer: clinical and molecular evidence. *Int J Mol Sci.* 2023;24(17):13501. doi:10.3390/ijms241713501
58. DeCosse JJ, Miller HH, Lesser ML. Effect of wheat fiber and vitamins C and E on rectal polyps in patients with familial adenomatous polyposis. *J Natl Cancer Inst.* 1989;81:1290-1297.
59. Aune D, Chan DS, Lau R, et al. Dietary fibre, whole grains, and risk of colorectal cancer: systematic review and dose-response meta-analysis of prospective studies. *BMJ.* 2011;343:d6617.
60. Celiberto F, Aloisio A, Girardi B, et al. Fibres and colorectal cancer: clinical and molecular evidence. *Int J Mol Sci.* 2023;24(17):13501. doi:10.3390/ijms241713501
61. Jacobs ET, Ahnen DJ, Ashbeck EL, et al. Association between body mass index and colorectal neoplasia at follow-up colonoscopy: a pooling study. *Am J Epidemiol.* 2009;169:657-666.
62. Wynder EL, Kajitani T, Kuno J, Lucas JC Jr, DePalo A, Farrow J. A comparison of survival rates between American and Japanese patients with breast cancer. *Surg Gynecol Obstet.* 1963;117:196-200.
63. Verreault R, Brisson J, Deschenes L, Naud F, Meyer F, Belanger L. Dietary fat in relation to prognostic indicators in breast cancer. *J Natl Cancer Inst.* 1988;80:819-825.
64. Newman SC, Miller AB, Howe CR. A study of the effect of weight and dietary fat on breast cancer survival time. *Am J Epidemiol.* 1986;123:767-774.
65. Campbell TM, Campbell EK, Culakova E, et al. A whole-food, plant-based randomized controlled trial in metastatic breast cancer: weight, cardiometabolic, and hormonal outcomes. *Breast Cancer Res Treat.* Published online March 6, 2024. doi:10.1007/s10549-024-07266-1
66. Holm LE, Callmer E, Hjalmar ML, Lidbrink E, Nilsson B, Skoog L. Dietary habits and prognostic factors in breast cancer. *J Natl Cancer Inst.* 1989;81:1218-1223.
67. Rock CL, Doyle C, Demark-Wahnefried W, et al. Nutrition and physical activity guidelines for cancer survivors. *CA Cancer J Clin.* 2012;62:243-274.
68. Tao MH, Shu XO, Ruan ZX, Gao YT, Zheng W. Association of overweight with breast cancer survival. *Am J Epidemiol.* 2006;163:101-107.
69. Hurtado-Barroso S, Trius-Soler M, Lamuela-Raventós RM, Zamora-Ros R. Vegetable and fruit consumption and prognosis among cancer survivors: a systematic review and meta-analysis of cohort studies. *Adv Nutr.* 2020;11(6):1569-1582. doi:10.1093/advances/nmaa082
70. Fairfield KM, Willett WC, Rosner BA, Manson JE, Speizer FE, Hankinson SE. Obesity, weight gain, and ovarian cancer. *Obstet Gynecol.* 2002;100:288-296.
71. Reeves GK, Pirie K, Beral V, et al. Cancer incidence and mortality in relation to body mass index in the Million Women Study: cohort study. *BMJ.* 2007;335:1134.
72. Breslow N, Chan CW, Dhom G, et al. Latent carcinoma of prostate at autopsy in seven areas. *Int J Cancer.* 1977;20:680-688.
73. Berkow SE, Barnard ND, Saxe GA, Ankerberg-Nobis T. Diet and survival after prostate cancer diagnosis. *Nutr Rev.* 2007;65:391-403.
74. Loeb S, Fu BC, Bauer SR, et al. Association of plant-based diet index with prostate cancer risk. *Am J Clin Nutr.* 2022;115(3):662-670. doi:10.1093/ajcn/nqab365
75. Ornish D, Weidner G, Fair WR, et al. Intensive lifestyle changes may affect the progression of prostate cancer. *J Urol.* 2005;174:1065-1069; discussion 1069-1070.
76. Earnest DL, Sampliner RE, Roe DJ, et al. Progress report: the Arizona phase III study of the effect of wheat bran fiber on recurrence of adenomatous colon polyps. *Am J Med.* 1999;106(1A):43S-45S. doi:10.1016/s0002-9343(99)00003-0



Recipes

The following recipes are from *The Cancer Survivor's Guide: Foods That Help You Fight Back!*
You'll be surprised how simple, delicious, and nutritious they are!

Roasted Red Pepper Hummus

Makes about 2 cups (8 1/4-cup servings)

Spread hummus on whole-wheat pita bread or serve as a dip for vegetables. This version is lower in fat than most commercial varieties, which is important when it comes to reducing hormone-dependent cancer risk and keeping your immune system operating smoothly.

1 15-ounce can garbanzo beans, or 1 1/2 cups cooked garbanzo beans
1 tablespoon tahini (sesame seed butter)
1/4 cup lemon juice
3 green onions, chopped
1 tablespoon chopped garlic (about 3 cloves)
1 teaspoon ground cumin
1/2 teaspoon black pepper
1/2 cup roasted red peppers, packed in water

If using canned garbanzo beans, drain them, reserving liquid, and rinse beans. Place beans, tahini, lemon juice, green onions, garlic, cumin, black pepper, and roasted peppers in food processor or blender and process until smooth. Add reserved bean liquid, or if using cooked beans, water or vegetable broth, as needed for a smoother consistency.

Per 1/4-cup serving: 80 calories; 2.1 g fat; 0.3 g saturated fat; 23.3% calories from fat; 0 mg cholesterol; 3.9 g protein; 12.5 g carbohydrate; 1.4 g sugar; 2.8 g fiber; 32 mg sodium; 36 mg calcium; 1.6 mg iron; 23.5 mg vitamin C; 299 mcg beta-carotene; 0.4 mg vitamin E

Recipe by Jennifer Reilly, RD

Cucumber, Mango, and Spinach Salad

Makes 10 to 12 servings

This salad boasts beta-carotene from the mango and lutein from the spinach, which are cousins in the carotenoid family and important antioxidants. Aside from cancer prevention and survival, lutein is recognized for its key role in eye health.

1 bag or bunch fresh spinach
1 mango, peeled and cut into bite-size pieces
1 large cucumber, peeled and sliced
6 green onions, thinly sliced
1/2 cup chopped fresh basil
Juice of 1 lime
1/2 cup seasoned rice vinegar
Freshly ground black pepper, or to taste

Wash and drain spinach, tear into bite-size pieces, if necessary, and put into a large serving bowl. Toss mango, cucumber, green onions, and basil in a medium bowl. Add lime juice and vinegar and stir to mix. Arrange mango mixture on spinach and sprinkle with black pepper.

Per serving (1/10 of recipe): 45 calories; 0.3 g fat; 0 g saturated fat; 5.5% calories from fat; 0 mg cholesterol; 1.5 g protein; 10.9 g carbohydrate; 7.4 g sugar; 1.7 g fiber; 219 mg sodium; 50 mg calcium; 1.3 mg iron; 19.1 mg vitamin C; 2,134 mcg beta-carotene; 1 mg vitamin E

Recipe by Amy Lanou, PhD

Any Veggie Coconut Curry

Makes 6 servings

The vegetables listed below are favorites, but any vegetables you have on hand can be used along with the coconut milk and spices. If you decide to use coconut milk, it is important to note that this product has a high fat content and should be used in moderation. Try lite coconut to get the sweet nutty coconut flavor without all the fat. Cauliflower, squash, and sweet potatoes are other tasty choices. This is a great way to use up “going-bad” fresh or frozen vegetables! Serve over rice or your favorite whole grain.

1 cup dry (uncooked) brown rice
2 cups water
1 large onion, sliced
4 garlic cloves, chopped (about 4 teaspoons)
3 large carrots, chopped
1/4 cup vegetable broth or water
1 1/2 tablespoons curry powder
1 teaspoon ground cumin
1/2 teaspoon turmeric
Pinch cayenne pepper
1 medium potato, chopped (peeling optional)
3 cups chopped kale



2 cups chopped broccoli florets, or 1 10-ounce package frozen chopped broccoli florets
 8 mushrooms, sliced
 1 15-ounce can garbanzo beans, drained and rinsed, or 1 1/2 cups cooked garbanzo beans
 1 cup fresh or frozen green peas
 1 cup lite coconut milk, or 1 cup nondairy milk plus 1 teaspoon coconut extract
 3 tablespoons reduced-sodium soy sauce

Bring rice and water to a boil in a medium saucepan. Lower heat and simmer, covered, until all the water is absorbed (about 30 minutes).

In a large saucepan, saute onion, garlic, and carrots in vegetable broth or water on medium-high heat until onion becomes translucent. Add curry, cumin, turmeric, and cayenne. Cook for 2 to 4 minutes, stirring often. Add potato, kale, broccoli, mushrooms, beans, peas, and coconut milk or nondairy milk mixture. Cover and reduce heat to medium-low. Simmer for 10 to 20 minutes, stirring occasionally, until potato can be pierced easily with a fork. Sprinkle with soy sauce before serving.

Per serving (1/6 of recipe): 320 calories; 5.2 g fat; 2.5 g saturated fat; 14.7% calories from fat; 0 mg cholesterol; 12.3 g protein; 59.3 g carbohydrate; 5.9 g sugar; 12.3 g fiber; 407 mg sodium; 115 mg calcium; 4.7 mg iron; 37.7 mg vitamin C; 6,312 mcg beta-carotene; 1.7 mg vitamin E

Recipe by Jennifer Reilly, RD

Colorful Corn Salsa

Makes 4 1/2 cups (18 1/4-cup servings)

Corn adds fiber, onions supply allyl sulfides, and tomatoes bring lycopene to this nutrient-rich salsa—all are cancer-fighting ingredients. Enjoy it with baked chips, stuffed in a burrito, or atop a bed of fresh greens.

1 cup fresh or frozen corn kernels
 2 medium tomatoes, chopped
 1/4 cup chopped red onion (about 1/2 medium red onion)
 1/4 cup chopped green bell pepper (about 1/2 medium bell pepper)
 1/4 cup chopped orange bell pepper (about 1/2 medium bell pepper)
 10 fresh basil leaves, chopped
 Juice of 1 lime
 3 tablespoons rice or cider vinegar

If using fresh corn, blanch in boiling water for 3 minutes and rinse in cold water. If using frozen corn that isn't thawed completely, either blanch it in boiling water for 2 minutes and drain, or microwave until thawed.

In a large bowl, combine all ingredients and set aside for 15 to 20 minutes to allow the flavors to develop. Serve at room temperature.

Per 1/4-cup serving: 13 calories; 0.1 g fat; 0 g saturated fat; 7.5% calories from fat; 0 mg cholesterol; 0.4 g protein; 3 g carbohydrate; 1.1 g sugar; 0.5 g fiber; 1 mg sodium; 4 mg calcium; 0.1 mg iron; 13.6 mg vitamin C; 88 mcg beta-carotene; 0.1 mg vitamin E

Recipe by Jennifer Reilly, RD



Low-Fat Guacamole

Makes 2 1/2 cups (10 1/4-cup servings)

The peas in this guacamole help to lower the fat content, and they are also rich in cancer-fighting fiber. Fiber helps your body get rid of excess cholesterol and cancer-causing compounds that are otherwise reabsorbed back into your bloodstream. Despite the addition of peas, this guacamole derives a good portion of its calories from fat. However, the total fat grams and calories are still quite low, and when eaten with baked chips and as part of a low-fat, plant-based diet, overall fat intake will be within the recommended range.

1 cup drained and rinsed canned green peas, or 1 cup fresh or frozen green peas
 1 ripe avocado
 1/2 cup mild salsa (commercial variety, or the Colorful Corn Salsa recipe on the left)
 1 garlic clove, finely chopped, or 1 teaspoon chopped garlic
 1 green onion, chopped (optional)
 Juice of 1 lemon
 1/2 teaspoon ground cumin
 1 tablespoon chopped fresh cilantro (optional)
 1/4 teaspoon salt, or to taste
 1/4 teaspoon black pepper, or to taste

If using fresh or frozen peas, blanch peas by placing them in boiling water for 2 minutes to soften, and then cool with cold water and drain. Peel avocado and cut into large chunks. Mash avocado and peas together using a potato masher or fork, or, if a very creamy texture is desired, in a food processor. Mix in salsa, garlic, green onion (if using), lemon juice, cumin, and cilantro (if using). Add salt and black pepper or to taste.

Per 1/4-cup serving: 45 calories; 2.7 g fat; 0.4 g saturated fat; 53.5% calories from fat; 0 mg cholesterol; 1.3 g protein; 4.9 g carbohydrate; 1.3 g sugar; 2.1 g fiber; 227 mg sodium; 12 mg calcium; 0.5 mg iron; 6.1 mg vitamin C; 118 mcg beta-carotene; 0.5 mg vitamin E

Recipe by Amy Lanou, PhD



395 mg sodium; 96 mg calcium; 3.9 mg iron; 16.8 mg vitamin C; 328 mcg beta-carotene; 1.3 mg vitamin E

Recipe by Jennifer Raymond, MS, RD

Easy Bean Salad

Makes about 10 1-cup servings

The simplicity and widespread enjoyment of this salad have made it a Food for Life classic. Plus, it has tons of fiber to help move carcinogens, and excess cholesterol and hormones out of your body to improve overall health.

- 1/2 cup low-fat Italian salad dressing
- 1 15-ounce can kidney beans, drained and rinsed, or 1 1/2 cups cooked kidney beans
- 1 15-ounce can pinto beans, drained and rinsed, or 1 1/2 cups cooked pinto beans
- 1 15-ounce can black-eyed peas, drained and rinsed, or 1 1/2 cups cooked peas
- 1 10-ounce package frozen lima beans (preferably Fordhook lima beans), thawed completely, 1 1/2 cups cooked lima beans, or 1 1/2 cups cooked green soybeans (shelled edamame)
- 1 cup frozen corn, thawed completely, or cooked fresh corn, chilled
- 1 large red bell pepper, seeded and chopped
- 1/2 medium purple onion, chopped
- 1 teaspoon salt, or to taste
- 1 teaspoon black pepper, or to taste

Toss all ingredients together. Serve cold or at room temperature. May be covered and stored in refrigerator for several days.

Per serving (1/10 of recipe): 183 calories; 3 g fat; 0.5 g saturated fat; 14.6% calories from fat; 0 mg cholesterol; 9.9 g protein; 31 g carbohydrate; 2.9 g sugar; 8 g fiber; 539 mg sodium; 43 mg calcium; 2.7 mg iron; 36.7 mg vitamin C; 311 mcg beta-carotene; 0.8 mg vitamin E

Recipe by Jennifer Reilly, RD

Penne With Kale, Tomatoes, and Olives

Makes 4 servings

The kale in this flavorful combination provides highly absorbable calcium and isothiocyanates, which have strong anticancer effects.

- 1 medium onion, chopped
- 1/4 cup vegetable broth or water
- 1 bunch kale, cut or torn into approximately 1-inch pieces (about 5 cups chopped)
- 2 14.5-ounce cans chopped, preferably fire-roasted, tomatoes, undrained, or 3 cups of freshly chopped tomatoes plus 1/2 cup water or vegetable broth
- 1/2 cup pitted and sliced Kalamata olives
- 1 tablespoon chopped fresh parsley
- 8 ounces dry whole-wheat penne pasta
- 1/4 cup dairy-free (vegan) Parmesan cheese substitute or nutritional yeast (optional)

Saute onion with vegetable broth or water over medium heat for 3 minutes. Add kale and tomatoes and their liquid. Bring to a boil and then reduce heat, cover, and simmer for 20 minutes.

Three-Bean Chili

Makes about 8 1-cup servings

This multicolor chili takes just 30 minutes to prepare, and it's chock-full of fiber, which not only enhances immune function but also rids the body of excess circulating hormones and carcinogens, thus lowering cancer risk. Serve it with brown rice or warmed tortillas and a green salad.

- 2 cups water, divided (1/2 cup plus 1/2 cup plus 1 cup)
- 1 large onion, chopped (about 2 cups)
- 1 teaspoon cumin seeds
- 6 garlic cloves, minced or pressed
- 1 green bell pepper, seeded and chopped
- 1 cup crushed tomatoes or tomato sauce
- 2 tablespoons chili powder
- 1 15-ounce can black beans, undrained, or 1 1/2 cups of cooked black beans with 1/2 cup of added water or vegetable broth
- 1 15-ounce can great northern beans, undrained, or 1 1/2 cups of cooked northern beans with 1/2 cup of added water or vegetable broth
- 1 15-ounce can red beans, undrained, or 1 1/2 cups of cooked red beans with 1/2 cup of added water or vegetable broth

Heat 1/2 cup water in a large pot. Add onion and cumin seeds and cook over high heat, stirring often, until onion is soft, 3 to 5 minutes. Add a bit more water if onion begins to stick.

Stir in garlic, bell pepper, and 1/2 cup water. Reduce heat to medium and cook 3 minutes, stirring occasionally.

Add tomatoes or tomato sauce, chili powder, and remaining 1 cup water. Cover and simmer 5 minutes.

Add beans and their liquid. Cover loosely and simmer 15 minutes.

Per 1-cup serving: 174 calories; 1 g fat; 0.2 g saturated fat; 5.1% calories from fat; 0 mg cholesterol; 10.8 g protein; 32.5 g carbohydrate; 3.7 g sugar; 8.3 g fiber;

Add olives and parsley. Cook an additional 5 minutes.

Meanwhile, cook pasta according to package directions. Drain and transfer to a serving bowl. Add kale mixture and toss gently. Serve immediately. Sprinkle vegan Parmesan substitute or nutritional yeast over top, if using.

Per serving (1/4 of recipe): 281 calories; 3.3 g fat; 0.5 g saturated fat; 10.5% calories from fat; 0 mg cholesterol; 12 g protein; 57.6 g carbohydrate; 8.3 g sugar; 8.7 g fiber; 497 mg sodium; 166 mg calcium; 5 mg iron; 53.9 mg vitamin C; 6,796 mcg beta-carotene; 2.9 mg vitamin E

Recipe by Amy Lanou, PhD

Berry Applesauce

Makes 4 1/2-cup servings

Serve this applesauce hot or cold. Berries give this applesauce a deep red or purple color and add a hefty dose of anthocyanins, potent cancer-fighting antioxidants.

2 cups peeled, cored, and chopped apples
2 cups fresh or frozen (unsweetened) strawberries, blueberries, or raspberries
1/2 cup frozen apple juice concentrate
1 teaspoon cinnamon

Combine all ingredients in a medium saucepan. Bring to a simmer, and then cover and cook over very low heat for about 25 minutes, or until apples are tender when pierced with a fork. Mash lightly with a potato masher or puree in a food processor, if desired.

Per 1/2-cup serving: 108 calories; 0.4 g fat; 0 g saturated fat; 3.5% calories from fat; 0 mg cholesterol; 0.8 g protein; 26.9 g carbohydrate; 20.1 g sugar; 2.7 g fiber; 11 mg sodium; 29 mg calcium; 0.9 mg iron; 49.2 mg vitamin C; 13 mcg beta-carotene; 0.4 mg vitamin E

Recipe by Jennifer Raymond, MS, RD

Raspberry Fudgy Brownies

Makes 16 servings

A brownie made with beans? You bet! The beans add fiber, calcium, and protein, making these brownies a nutrition indulgence.

2 15-ounce cans low-sodium black beans, drained and rinsed
1 cup pitted dates
1 cup all-fruit raspberry jam
2 teaspoons pure vanilla extract
1/4 cup plus 2 tablespoons whole-wheat pastry flour
1 cup unsweetened cocoa powder
1/4 teaspoon sea salt
1/2 cup mini chocolate chips or 1/2 cup raspberry all-fruit jam, for topping

Preheat the oven to 350 F. Line an 8-by-8-inch baking pan with parchment paper. Combine the black beans, dates, jam, and vanilla in a food processor. Blend until smooth. Add the flour, cocoa powder, and sea salt and blend again. Pour into the prepared pan and smooth the top with moist hands. If desired, sprinkle evenly with chocolate chips or spread with the raspberry jam.

Bake for 30 minutes. Remove from the oven and cool completely. Use the parchment paper to lift the brownies out of the pan. Cut into 16 squares. Refrigerate for up to 1 week, stored in a covered container.

Per serving (1/16 of recipe): 205 calories; 2.8 g fat; 1.4 g saturated fat; 11% calories from fat; 0 mg cholesterol; 6 g protein; 43 g carbohydrate; 18.8 g sugar; 9.1 g fiber; 359 mg sodium; 54 mg calcium; 2.5 mg iron; 2.3 mg vitamin C; 1 mcg beta-carotene; 0.1 mg vitamin E

Recipe by Christine Waltermeyer

Mixed Berry Banana Smoothie

Makes 4 servings

Frozen berries in smoothies add fiber, taste, and iciness, not to mention the hefty dose of cancer-fighting compounds!

1 banana
1 cup frozen mixed berries (raspberries, blueberries, strawberries)
1 cup fortified vanilla nondairy milk
2 tablespoons maple syrup (optional)
2 tablespoons calcium-fortified orange juice concentrate

Place all ingredients in a blender. Blend at high speed until smooth, stopping the blender occasionally to move the unblended fruit to the center with a spatula.

Per serving (1/4 of recipe): 107 calories; 1.4 g fat; 0.2 g saturated fat; 11.5% calories from fat; 0 mg cholesterol; 3.3 g protein; 22.3 g carbohydrate; 12.9 g sugar; 3.7 g fiber; 36 mg sodium; 130 mg calcium; 1 mg iron; 33.1 mg vitamin C; 24 mcg beta-carotene; 1.3 mg vitamin E

Recipe by Amy Lanou, PhD



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About the Physicians Committee for Responsible Medicine

With more than 2 million people predicted to be diagnosed with cancer in the United States each year—and many more cases in other countries across the globe—there is an urgent need for a new direction in battling this disease. The Physicians Committee for Responsible Medicine has created materials and programs to educate the public on how a healthful diet can protect us from cancer and help us regain our health once cancer has been diagnosed.

The Physicians Committee also conducts clinical research, hosts national conferences, and provides literature and resources to thousands of health professionals. The organization's longest-running nutrition education program is the Food for Life nutrition and cooking class program. Since 2001, it has helped tens of thousands of individuals around the world, including those looking to prevent or survive cancer, understand the power of their plates for optimal health. The classes provide nutrition education and serve as support groups. *The Cancer Survivor's Guide: Foods That Help You Fight Back* can also provide home access to this powerful nutrition message.